Assume/Guarantee Contract of `GyroPick`

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1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet :
wa, wa_out′, wal, wal′, countPick, countPick′, waThr, waThr′, pickThr, pickThr′, clock, clock′

Inputs :

- `wa` $\mapsto$ wa : $\mathbb{N} \rightharpoonup \mathbb{R}$

Outputs :

- `wa_out` $\mapsto$ wa_out : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs : None

States :

- `wal` $\mapsto$ wal : $\mathbb{N} \rightharpoonup \mathbb{R}$
- `countPick` $\mapsto$ countPick : $\mathbb{N} \rightharpoonup \mathbb{N}$

Parameters :

- `waThr` $\mapsto$ waThr : $\mathbb{R}$
- `pickThr` $\mapsto$ pickThr : $\mathbb{N}$

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$\begin{aligned} Assumptions \triangleq\ & \mathsf{len}(\mathsf{wa}) \geq 9 \\ & \wedge\ \mathsf{len}(\mathsf{wal}) = \mathsf{len}(\mathsf{countPick}) = 9 \end{aligned}$$

Guarantees :

$$\begin{aligned} Guarantees \triangleq\ & Check_1 \wedge Check_2 \\ Check_1 \triangleq\ & \mathsf{len}(\mathsf{wa_out}') = \mathsf{len}(\mathsf{wal}') = \mathsf{len}(\mathsf{countPick}') = 9 \\ & \wedge\ \mathsf{waThr}' = \mathsf{waThr} \\ & \wedge\ \mathsf{pickThr}' = \mathsf{pickThr} \\ & \wedge\ \mathsf{clock}' = \mathsf{clock} \\ Check_2 \triangleq\ & \forall 0 \leq i < 9 \cdot \begin{pmatrix} \mathsf{if}\ |\mathsf{wa}[i] - \mathsf{wal}[i]| > \mathsf{waThr}\ \wedge\ \mathsf{countPick}[i] + 1 < \mathsf{pickThr}\ \mathsf{then} \\ \quad \mathsf{wa_out}'[i] = \mathsf{wal}'[i] = \mathsf{wal}[i] \\ \quad \wedge\ \mathsf{countPick}'[i] = \mathsf{countPick}[i] + 1 \\ \mathsf{else} \\ \quad \mathsf{wa_out}'[i] = \mathsf{wal}'[i] = \mathsf{wa}[i] \\ \quad \wedge\ \mathsf{countPick}'[i] = 0 \\ \mathsf{endif} \end{pmatrix} \end{aligned}$$